MIL-C-45461 (Ord)
29 January 1960

MILITARY SPECIFICATION

CHARGE, PROPELLING FOR
ROCKET MOTOR, M4

1. SCOPE

1.1 This specification covers the inspection and testing for one type propelling charge designated as Charge, Propelling, for Rocket Motor M4.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition
MIL-P-45460 (Ord) - Propellant, T2

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-292 - Ballistic Nomenclature Rocket Static Tests
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

FSC 1375

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DRAWINGS

ORDNANCE CORPS

7548417 - Charge, Propelling
7548416 - Rocket Motor, M4, Assembly
76-1-1039 - Drum, Metal, for 75mm and 105mm Howitzer, Propelling Charges

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes
GL-8792441 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling charge.-The propelling charge shall comply with all requirements specified on Drawing (dwg.) 7548417, and with all requirements specified in applicable specifications.

3.3 Composition.-The composition of thepropellant shall be as specified in MIL-P-45460(Ord)The manufacturer shall use no ingredients other than those specified in MIL-P-45460 (Ord).

3.4 Straightness.-The propellent charge shall be straight within 0.010 inches as measured from the maximum allowable outside diameter.

3.5 Flatness and squareness.-The ends of the propellent charge shall be flat within 0.010 inches and square within 2 degrees (^{0}).

3.6 Static functioning.-The propelling charge shall not fail to ignite and shall comply with the following requirements when tested as specified in 4.3.1.

3.6.1 Action time.

3.6.1.1 At 0^{0} / 3^{0}Fahrenheit (F).-The action time shall not be less than 0.43 nor greater than 0.72 seconds (sec.).

3.6.1.2 At 120^{0} / 3^{0}F.-The action time shall not be less than 0.28 nor greater than 0.44 secs.

3.6.2 Maximum thrust (F)

3.6.2.1 At 0^{0} / 3^{0}F.-The F max. shall not exceed 600 pounds.

3.6.2.2 At 120^{0} / 3^{0}F.-The F max. shall not exceed 600 pounds.

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3.6.3 Total Impulse (I)

3.6.3.1 At 0° ∕ 3°F.-The total impulse shall not be less than 100 nor greater than 150 pound force seconds (lb. f-sec).

3.6.3.2 At 120° ∕ 3°F.-The total impulse shall not be less than 105 nor greater than 160 pound f-sec.

3.7 Workmanship.-The propelling charge shall be free of dirt, grease, and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The quality assurance provisions of this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD's-109 and 292 in order to define terms used herein. Inspection shall be performed in accordance with this specification, Specification MIL-G-2550 and other specifications referenced in any of the contractual documents.

4.1.1 Contractor inspection.-Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

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4.1.3 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of the agent's authority to bind his principal.

4.1.4 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied, and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with.

4.1.4.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and test are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.4.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.1.4.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect

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on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of propelling charges produced by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting performance, safety or fit, as determined by the Government, shall not be cause for changing the lot number. Each lot shall contain Propellant, T2 from one lot from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Rocket motor (see dwg. 7548417)

Categories and defects	Method of inspection
Critical	
a. Charge cracked or broken	Visual
Major AQL 0.25 percent	
1. Outside diameter	Gage
2. Length	Gage
Minor AQL 1.00 percent	
41. Straightness of charge (see 3.4)	Gage
42. Squareness of ends (see 3.5)	Gage

4.2.3 Testing.

4.2.3.1 Static functioning - major defect (see 3.6 and 6.3).

4.2.3.1.1 0° and 120°F.-The sampling procedure for this test shall be in accordance with Standard MIL-STD-414, section B, Table B-1, code letter F and an AQL of 0.65 percent for each limit and each temperature separately. A complete rocket motor consisting of two propelling charges shall be considered one sample.

4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 and 4.2.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment identified by Index of Inspection Equipment No. 879241, or approved

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equivalents thereto. In inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.-The tolerance specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.3.1 Static functioning.

4.3.1.1 0° ± 3°F.-The propelling charge shall be loaded into M4 rocket motor as shown on dwg. 7548416, or into a pressure vessel simulating the M4 rocket motor, and the motors conditioned at 0 ± 3°F for a period of not less than 18 hours. Within 5 minutes after removal from the conditioning chamber, the motors shall be tested in a suitably barricaded test stand. Initiation of the rocket motor ignition system shall be by use of the applicable fuze lighter which is standard for the item. The test shall provide a permanent record of the thrust developed by the propelling charge of each rocket motor against time.

4.3.1.2 At 120° ± 3°F.-The propellent charge shall be tested as specified in 4.3.1.1 except that the conditioning temperature shall be 120° ± 3°F.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level B - limited military pack.-The propelling charge shall be packed in accordance with dwg. 76-1-1039.

5.1.2 Level C - minimum military pack.-The propelling charge shall be packed as specified in 5.1.1.

5.2 Marking.-Marking shall be as specified on dwg. 76-1-1039.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Submission of test data.-In addition to the normal distribution of records when the item is procured by the Department of the Army, a copy shall be forwarded to the following:

1 copy - Commanding Officer
Picatinny Arsenal
Dover, New Jersey, ATTN: ORDEB-DE2

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6.3 Application.-When the manufacture of the propelling charge and the loading of the rocket motor, M4 assembly are performed by the same contractor, the tests specified in 4.2.3 of this specification need not be performed.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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